■ FINAL PHASE 1 DEFINITION

1. ■ Use Case (Locked)

Web-based symptom triage system for college students to assess common illnesses and decide next action

2. ■ Target User (Locked)

Age: 18–25

Tech-savvy

Mild to moderate symptoms

Wants quick decision (ignore vs doctor)

3. ■ User Flow (Precise)

Step-by-step system flow:

- User opens web app
- Selects symptoms from predefined list (multi-select)
- System validates:
- Minimum 2 symptoms required
- Invalid/empty input rejected
- Input converted into feature vector
- Model predicts top 3 diseases with probabilities
- Logic layer:
- Assigns risk level (Low / Medium / High)
- Filters unrealistic combinations
- Output displayed:
- Diseases + probability
- Risk level
- Explanation
- Recommended action
- User decides next step

4. ■ Inputs (Locked + Precise)

- Symptoms: predefined list (~40–60 symptoms)
- Example: fever, cough, fatigue, nausea
- Format: multi-select checklist
- Minimum input: 2 symptoms
- Optional (Phase 2):

- Duration
- Severity

5. ■ Outputs (Structured)

- Top 3 diseases
- Probability score (%)
- Risk level:
- Low → self-care
- Medium → consult doctor
- High → urgent care
- Explanation format:

“Prediction based on presence of: fever + cough + fatigue”

6. ■ Success Metrics (Sharpened)

- Model Accuracy ≥ 85%
- API response time ≤ 2 seconds
- Handles invalid input gracefully
- User can understand result in < 10 seconds

7. ■ Tech Stack (Decided — No More Switching)

- Backend → FastAPI
- ML → scikit-learn (Random Forest baseline)
- Frontend → React
- Database → PostgreSQL